

MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES
(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting – Inter-process Communication

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1. Define Interprocess Communication (IPC). Briefly explain its role in an operating system

Interprocess communication, or IPC, is how separate programs running on the same computer—or sometimes on different computers—share information and coordinate their work. The operating system provides IPC tools so that programs can communicate safely without directly accessing one another's memory.

2. Why is IPC necessary? Explain with suitable examples.

Programs often need to work together. For example, a browser might ask another process to play a video, or a print application might send a document to the computer's printing service. IPC lets them pass information and coordinate their actions.

3. Describe the different IPC mechanisms available in an operating system. Distinguish between mechanisms used for data exchange and those used primarily for synchronization or notification.

Some IPC methods mainly help programs exchange data, while others mainly help them coordinate or notify each other.

4. Compare the advantages and limitations of the IPC mechanisms discussed above.

Method	Advantages	Limitations
Pipes and named pipes	Simple and useful for passing data.	Pipes are often one-way; named pipes are generally limited to a single computer.
Message queues	Keep messages organized and allow processes to work at different times.	The operating system may limit how many messages or how much data can be stored.
Shared memory	Very fast, especially when processes need to share lots of data.	Processes must carefully coordinate access, or they may accidentally overwrite data.
Sockets	Can connect processes on one computer or across a network.	They require more setup, and network communication needs extra security.
Signals and events	Useful for quickly notifying a process.	They usually carry little information.
Semaphores and mutexes	Help prevent processes from interfering with each other when using shared resources.	If used incorrectly, processes can get stuck waiting for one another.

Pipes, queues, shared memory, and sockets are mainly used to pass data. Signals, events, and synchronization tools are mainly used to coordinate activity. Some methods can serve both purposes.

5. From a cybersecurity perspective, discuss the security risks associated with IPC and recommend measures to secure communication between processes.

IPC can create security problems if it isn't properly controlled. For example, an unauthorized process might read or change shared data, pretend to be a trusted program, or send a service harmful input. Too many messages could also overwhelm a process.

To make IPC safer, the operating system should restrict access to communication channels, and programs should check who is sending each request. They should also validate incoming data, limit message sizes, protect sensitive information, and use synchronization tools carefully. Monitoring errors and keeping software updated can help identify and prevent further problems.